

Expectation of Originality:

Faculty of Engineering and Computer Science Expectations of Originality This form sets out the requirements for originality for work submitted by students in the Faculty of Engineering and Computer Science. Submissions such as assignments, lab reports, project reports, computer programs and take-home exams must conform to the requirements stated on this form and to the Academic Code of Conduct. The course outline may stipulate additional requirements for the course.

1. Your submissions must be your own original work. Group submissions must be the original work of the students in the group.
2. Direct quotations must not exceed 5% of the content of a report, must be enclosed in quotation marks, and must be attributed to the source by a numerical reference citation¹. Note that engineering reports rarely contain direct quotations.
3. Material paraphrased or taken from a source must be attributed to the source by a numerical reference citation.
4. Text that is inserted from a web site must be enclosed in quotation marks and attributed to the web site by numerical reference citation.
5. Drawings, diagrams, photos, maps or other visual material taken from a source must be attributed to that source by a numerical reference citation.
6. No part of any assignment, lab report or project report submitted for this course can be submitted for any other course.
7. In preparing your submissions, the work of other past or present students cannot be consulted, used, copied, paraphrased or relied upon in any manner whatsoever.
8. Your submissions must consist entirely of your own or your group's ideas, observations, calculations, information and conclusions, except for statements attributed to sources by numerical citation.
9. Your submissions cannot be edited or revised by any other student.
10. For lab reports, the data must be obtained from your own or your lab group's experimental work.
11. For software, the code must be composed by you or by the group submitting the work, except for code that is attributed to its sources by numerical reference.

You must write one of the following statements on each piece of work that you submit: For individual work: "I certify that this submission is my original work and meets the Faculty's Expectations of Originality", with your signature, I.D. #, and the date. For group work: "We certify that this submission is the original work of members of the group and meets the Faculty's Expectations of Originality", with the signatures and I.D. #s of all the team members and the date. A signed copy of this form must be submitted to the instructor at the beginning of the semester in each course. I certify that I have read the requirements set out on this form, and that I am aware of these requirements. I certify that all the work I will submit for this course will comply with these requirements and with additional requirements stated in the course outline.

Course Number: COEN 313

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Signature: C . L .

Date: 22/04/2026

Red light green light project:

Design methodology and 7 segment displays

For my design I knew the simplest way to go was to have a mealy machine with red, light, idle and done states, where idle was the waiting between rounds state. I started the design with 1 player, and used some online resources to guide me on how to use the fpga's clock.

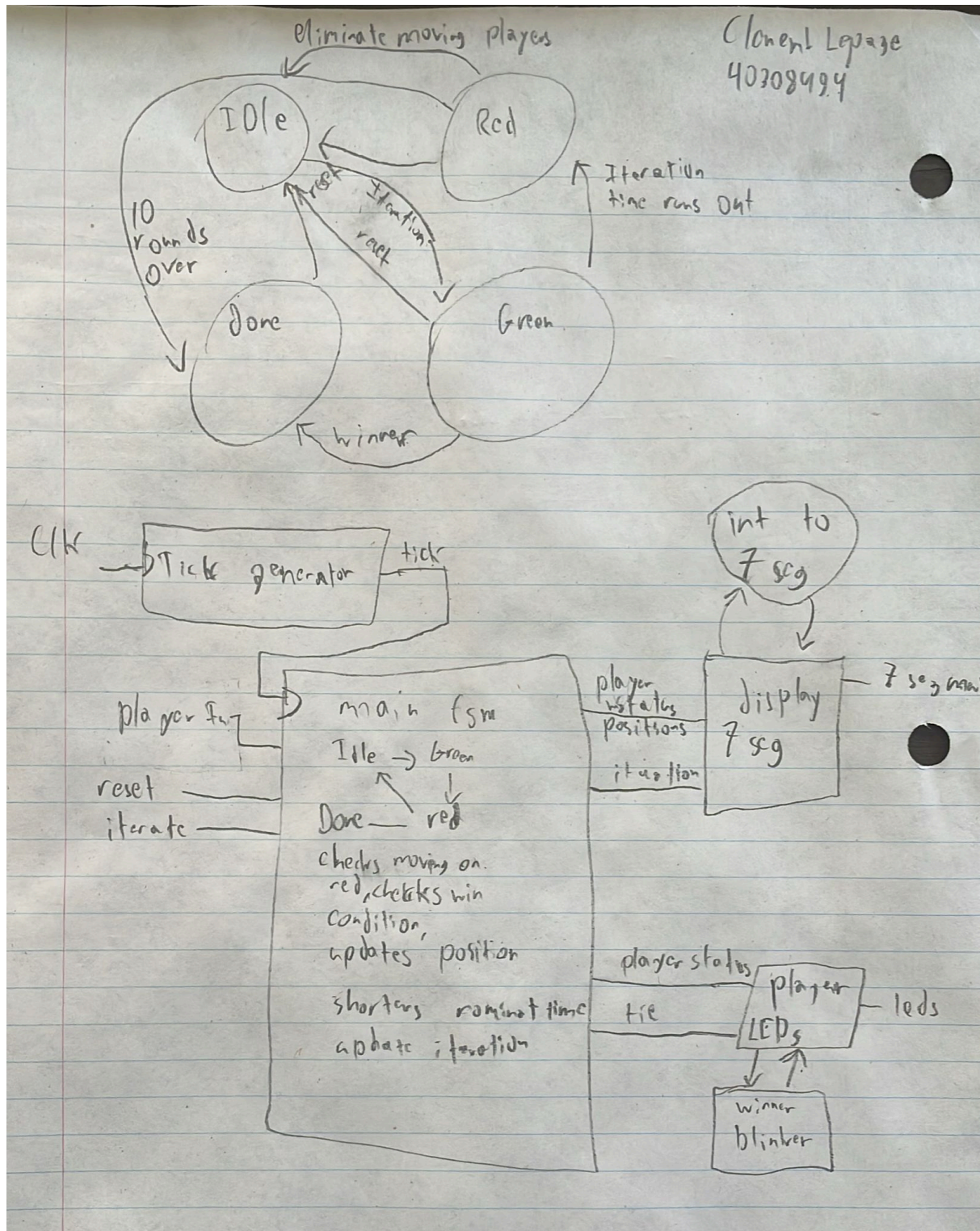
Needed online help for driving all the 7 segment displays using anode and cathode method, I also used a function I found online to convert an integer into 7 segment output. Even if functions were outside the scope of the course, this made sense to use and I understand how it works.

For the outputs, I chose to display all 4 players with 1 digit of precision on the 7 segment displays as I thought additional inputs to cycle through them would not make for great gameplay. 2 digits output time in seconds.tenths for how long is remaining in this iteration. I also included the iteration number between them, so the players can tell when the game is going to end. For iteration number (1-10) and player scores (0-12), I used hexadecimals to represent 10 and 11 (a and b).

I also chose to make all four players blink after 10 iterations with no winner as this is a tie, and didn't want confusion with all four players eliminated

Note: for testing the do file, the game progressed at 1 tick/clock cycle instead of the normal 1 tick /100,000 clock cycles on the 100Mhz fpga clock.

Conceptual block diagram:



Resources used:

<https://www.fpga4student.com/2017/09/vhdl-code-for-seven-segment-display.html>

<https://www.fpga4student.com/2016/11/vhdl-code-for-digital-clock-on-fpga.html>

<https://github.com/Digilent/digilent-xdc/blob/master/Nexys-A7-100T-Master.xdc>